

$$\begin{aligned}
L^a &= J^{ab}\omega_b \\
J^{ii} &= \int x^{j^2} - x^{k^2} dm(j, k \neq i) \quad dm = \rho dV = \frac{m}{a^3} dx dy dz \\
J^{11} &= \int_V y^2 + z^2 dm \int y^2 + z^2 \rho dx dy dz = ay^2 + az^2 dy dz = \frac{1}{3} a * a^3 + a^2 z^2 dz = \\
&\frac{1}{3} a^2 * a^3 + \frac{1}{3} a^2 a^3 = \frac{2}{3} a^5 \\
J^{11} &= \frac{m}{a^3} \frac{2}{3} a^5 = \frac{2}{3} ma^2 \\
J^{ii} &= \frac{2}{3} ma^2 \\
J^{12} &= - \int x y \rho dx dy dz = -\frac{1}{2} a^2 \rho y dy dz = -\frac{1}{4} a^4 \rho dz = -\frac{1}{4} a^5 \rho = -\frac{1}{4} ma^2 \\
J^{\nu\mu} &= -\frac{1}{4} ma^2 \\
L^a &= J_b^a \omega^b; \omega^b = \frac{1}{\sqrt{3}} (i^b + j^b + k^b) \\
L^a &= \frac{1}{\sqrt{3}} (J^{a1} + J^{a2} + J^{a3}) = \frac{1}{\sqrt{3}} ((\frac{2}{3} ma^2 - \frac{1}{2} ma^2) i^a + (\frac{2}{3} ma^2 - \frac{1}{2} ma^2) j^a + \\
&(\frac{2}{3} ma^2 - \frac{1}{2} ma^2) k^a) \\
&= \frac{1}{\sqrt{3}} (\frac{1}{6} (j^a + j^a + k^a)) ma^2 \\
T &= \frac{1}{2} \omega^a J_{ab} \omega^b = \frac{1}{2} \frac{1}{\sqrt{3}} (i^b + j^b + k^b) \frac{1}{\sqrt{3}} (\frac{1}{6} (j^a + j^a + k^a)) ma^2 = \frac{1}{12} ma^2 \\
&\text{answer(1)} \\
&\frac{1}{\sqrt{3}} (\frac{1}{6} (j^a + j^a + k^a)) ma^2 \tag{1}
\end{aligned}$$

answer(2)

$$T = \frac{1}{12} ma^2 \tag{2}$$

$$\begin{aligned}
m \frac{dv}{dt} &= -\frac{1}{2} v \quad m \frac{d \ln v}{dt} = -\frac{1}{2} \ln v = C - \frac{t}{2m} ; v = C e^{-\frac{t}{2m}} ; \frac{1}{2} m v^2 = mgR \\
v &= \sqrt{2gR} \\
v|_{t=0} &= C = \sqrt{(2gR)} ; v = \sqrt{(2gR)} e^{-\frac{t}{2m}}
\end{aligned}$$

1 Oct 22, example

1.1 metter = m stick. And with a m ball.

$$\begin{aligned}
\frac{1}{2} \frac{1}{3} m l^2 \omega^2 &= \frac{1}{2} m g l; \omega = \sqrt{\frac{3g}{l}} \\
\frac{1}{3} m l^2 \omega &= \frac{1}{3} m l^2 \omega_0 + m l^2 \omega_0 4; \omega_0 = \frac{1}{4} \omega \\
2 m g \frac{3}{4} l (1 - \cos \theta) &= \frac{1}{2} \frac{1}{3} m l^2 \omega_0^2 + \frac{1}{2} m l^2 \omega_0^2 = \frac{2}{3} m l^2 \omega_0^2 = \frac{2}{3} m l^2 \frac{3g}{16l} = \frac{1}{8} m g l \\
\frac{1}{12} &= (1 - \cos \theta); \cos \theta = \frac{11}{12}
\end{aligned}$$

1.2 a stick m and 60 degree.

$$\begin{aligned}
L &= \frac{1}{3} m l^2 \\
\frac{1}{2} \frac{1}{3} m l^2 \omega^2 &= m g \frac{1}{2} l \sin 30; \omega = \sqrt{\frac{3g}{2l}} \\
m g \cos 60 - T_1 &= -m \frac{1}{2} l \omega^2; T_1 = \frac{1}{2} m g + \frac{1}{2} m l \frac{3g}{2l} = \frac{5}{4} m g \\
\frac{1}{3} m l^2 \beta &= \frac{1}{2} m g l \sin 60; \beta = \frac{3\sqrt{3}g}{4l} \\
a_c &= \beta \frac{1}{2} l = g \sin 60 + T_0/m; T_0 = \frac{\sqrt{3}}{8} m g
\end{aligned}$$

$$\begin{aligned}
|T| &= \sqrt{\left(\frac{103}{64}\right)}mg \\
\text{way2: } a &= a' + a_c + 2v' \times \omega + r' \times \dot{\omega} + r' \times \omega \times \omega \\
r &= r' + r_c; v = v' + v_c + r' \times \omega; a = a' + a_c + v' \times \omega + v' \times \omega + r' \times \dot{\omega} + r' \times \omega \times \omega \\
a' &= a - a_c - 2v' \times \omega - r' \times \dot{\omega} - r' \times \omega \times \omega \\
r' \times a' &= -r' \times a_c - r' \times r' \times \dot{\omega} = r' \times \beta' \times r' = -\frac{1}{2}l^2\beta = \frac{1}{2}l^2\frac{3\sqrt{3}g}{4l}; \frac{3\sqrt{3}gl}{8}\frac{1}{12}ml^2 = \\
\frac{1}{4}l^2F_n; F_n &= \frac{\sqrt{3}}{8}mg \\
\frac{3\sqrt{3}mgl}{8} \\
\frac{1}{12}ml^2\frac{\sqrt{3}gl}{8} &= \frac{1}{2}l \times F_n
\end{aligned}$$

1.3 Oct 27

the ball have angular velocity, and floor have friction force:

$$\begin{aligned}
&\text{way 1:} \\
&I\omega_0 - I\omega = \mu mg R \Delta t \\
&m\omega R = \mu mg \Delta t \\
&\text{so we get } \omega = \frac{2}{7}\omega_0 \\
&\text{and the } \Delta t = \frac{\frac{2}{7}\omega_0 R}{\mu g} \\
&\text{so } x = \frac{1}{2}\mu g t^2 = \frac{2}{49\mu g}\omega_0^2 R^2 \\
&\text{way2:} \\
&T = \frac{1}{2}m\dot{x}^2 + \frac{1}{5}mR^2\dot{\theta}^2 \\
&v = 0; W = \mu mgx - \mu mgR\theta \\
&L = \frac{1}{2}m\dot{x}^2 + \frac{1}{5}mR^2\dot{\theta}^2 + \mu mgx - \mu mgR\theta \\
&m\ddot{x} = \mu mg; \frac{2}{5}mR^2\ddot{\theta} = -\mu mgR \\
&m\Delta\dot{x} = \mu mg\Delta t; \frac{2}{5}mR^2\Delta\dot{\theta} = -\mu mgR\Delta t \\
&mv = \mu mg\Delta t; \frac{2}{5}mR^2(\omega - \omega_0) = -\mu mgR\Delta t; \omega = vR \\
&m\omega R = \mu mg\Delta t; \frac{7}{5}mR^2\omega = \frac{2}{5}mR^2\omega_0; \omega = \frac{2}{7}\omega_0 \\
&\Delta t = \frac{2\omega_0 R}{7\mu g} \\
&\Delta x = \frac{1}{2}\mu g\left(\frac{2\omega_0 R}{7\mu g}\right)^2 = \frac{2\omega_0^2 R^2}{49\mu g}
\end{aligned}$$

1.4 Nov 10

3 Lagrangian questions

(A) a surface with a heavy object and a turntable.

$$L = \frac{1}{2}m_1\dot{l}^2 + \frac{1}{2}m_2\dot{l}^2 + \frac{1}{4}m_1\dot{l}^2 - (C - m_2gl)$$

$$\frac{3}{2}m_1\ddot{l} + m_2\ddot{l} = m_2g$$

$$\ddot{l} = \frac{m_2g}{\frac{3}{2}m_1 + m_2}$$

(B) a ramp with a turntable

$$L = \frac{3}{4}m\dot{l}^2 - (C - mgsin\theta l)$$

$$\frac{3}{2}m\ddot{l} = mgsin\theta$$

$$\ddot{l} = \frac{2}{3}gsin\theta$$

(C) A massive turntable with a heavy object

$$L = \frac{1}{2}m'\dot{l}^2 + \frac{1}{2}\frac{1}{2}m\dot{l}^2 - (C - m'gl)$$

$$m'\ddot{l} + \frac{1}{2}m\ddot{l} = m'g$$

$$\ddot{l} = \frac{m'g}{m' + \frac{1}{2}m}$$